Chained-transfer leads 2026-11-12

Chained-transfer leads, 2026-11-12: amortized IC, certificates, genus 3 at larger p

Owner request (2026-11-12): chase the open leads. The transfer is a one-time cost, because with a trapdoor it happens once, so it never enters a per-instance total. Three parallel experiments ran, and the integration session reproduced each from the final source (`repro/REPRO.md`). Every number is `[pcc:unverified]`.

1. Amortized index calculus (`amortized/AMORTIZED.md`)

The trapdoor argument applies to IC's own per-curve work too. Factor-base logs on `J(C)` are computed once, and each target then costs only a descent. The protocol was frozen before data. It compared against plain rho and against Bernstein–Lange (BL) precomputed rho matched by memory and by budget. That was the control that removed an earlier "IC ahead of rho online" reading here (`research/20261016-n83-precomputed-rho-control`).

- **Against plain rho:** the online cost per target is 2.0–3.1% of plain rho at all 11 primes from 211 to 205,661. At p = 1619 it is about 2,100× below the earlier IC that put the target into every relation (15.7 against 33,196 group ops) [pcc:unverified].
- **Against matched BL rho (the admissible comparison):** IC is 1.3–11% above it online in 10 of 11 rows. At p = 25,693 the two tie, and the paired interval contains 0. The frozen rule fails in 11 of 11 rows [pcc:unverified].
- **Why (integration analysis):** the factor base is at least as large as the target group in every row, with `fb/r` from 0.98 to 125 [pcc:unverified]. Genus-2 IC over `F_p` needs about `p/2` degree-1 places. The transferred elliptic subgroup has order `r ≤ #E ≈ p`. So the precomputation stores a log for as many places as the group has elements. With that much memory any method answers in a few operations; tabulating all `r` multiples of `G` would answer at once. The low online cost is a time–memory trade, which BL rho with the same memory reproduces. Amortizing pays only when the factor base is much smaller than the group. That holds for an attack on the whole Jacobian (about `p^g` elements), not for an elliptic subgroup of size about `p` pushed into it.
- The LA here is a weight-2 union-find, 26–148× cheaper than the dense count in `results_v2.json`. Pricing it either way leaves the verdict unchanged.

2. Certificates, oracle-free LA, real instance (`V3.md`)

These close replay findings F4, F5 and F14 [pcc:unverified]:

- **F4:** IC's elimination no longer checks against a BSGS answer.
- **F5:** the instance pair is `(φ^*(P), φ^*([x]P))` itself.
- **F14:** every solve emits a `chained.solve.certificate/v1`, and `verify_certs` passed 16 of 16 [pcc:unverified].

IC/rho is 8.0–40.8× at 0 GAE and 42–754× at 1 GAE. That is within 1.6 of v2, on the same curves and planted logs.

There is an open instrument question. On this path the LA mulmod counter equals `(fb + 2) × rows` in every case. That looks like a matrix-size count, not elimination work.

3. Genus 3 at larger p (`tower_large/TOWER_LARGE.md`)

A word-size Jacobian (`hyper64.rs`, `p < 2^31`) matched the `p < 256` implementation on every comparison at 8 primes before any larger-p number was taken [pcc:unverified]. Results from p = 251 to 64,007:

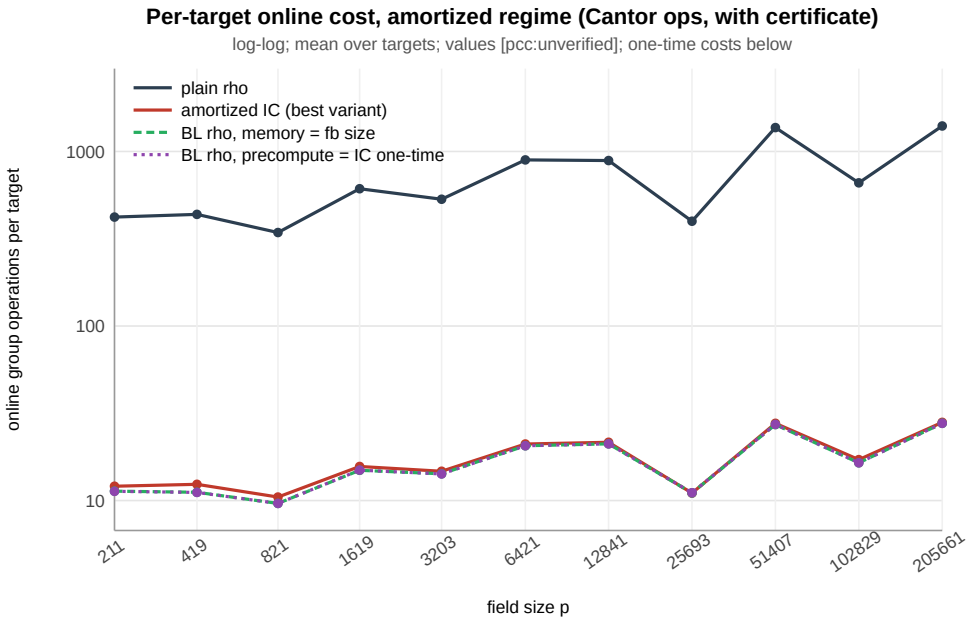
- IC3/IC2 is 2.92–3.10 for `p ≥ 503`.
- Trials per relation are 2.00 and 6.04 against `2!` and `3!`.
- IC group ops grow with slope 0.99 in p, and rho walk steps with slope 0.56.
- Genus 2 is 47× rho at p = 64,007 at 0 GAE.
- The only sub-rho cell is genus 2 at p = 251 at 0 GAE (0.67). That ordering flips inside the band.
- Genus-3 LA scales with slope 2.1 ($2.2 \cdot 10^9$ mulmods at p = 64,007), and rho on E is dominated by fixed set-up (slope 0.21).

Both instrument limits are listed below.

Next steps

1. **Endomorphism-degree census.** A non-integer endomorphism of degree `d` needs $x^2 + |D|y^2 = 4d$ with $y \neq 0$, so $d \geq |D|/4$. Count how often lab and ecdb curves have one of degree ≤ 11. That decides the reach of the small-`D_E` protocol (`research/small_de_endo_fb_20261112/PROTOCOL.md`).
2. **Volcano ascent to the maximal order.** `End(E)` varies within an isogeny class and `d_K` does not. Carry the target to the smallest-`|D|` curve in its class, counted as a one-time transfer, before endomorphism-adapted IC.
3. **The small-`D_E` experiment itself**, on curves step 1 shows exist.
4. **Cover mechanism test.** Push `J(C)` relations through `c_*` to `E` and measure the induced factor base. The bielliptic cover should reproduce 2-term IC on `E` over about half its points. That makes “the Prym does not help” a measurement, and it sets the bar a cover must clear.
5. **Instrument fixes.** Use structured Gaussian elimination plus Wiedemann with a counted LA, double-large-prime variation (needed for the `Õ(p^{2-2/g})` exponent at `g = 3`), and a rho reference of walk steps plus amortized set-up with the negation map.
6. **Validation gate.** Replay certificates, a validator report and `VALIDATION.json` for v3 and tower_large, plus a replay by an agent other than the producer.
7. **Proof note.** `J_C ~ E × P` and `c_*c^* = [deg c]`: every map out of `J_C` either kills the target subgroup or factors through `E`, and a smooth-order Prym simplifies only the complement. A Lean statement of the `c_*c^*` step would be a candidate for crypto-lean.

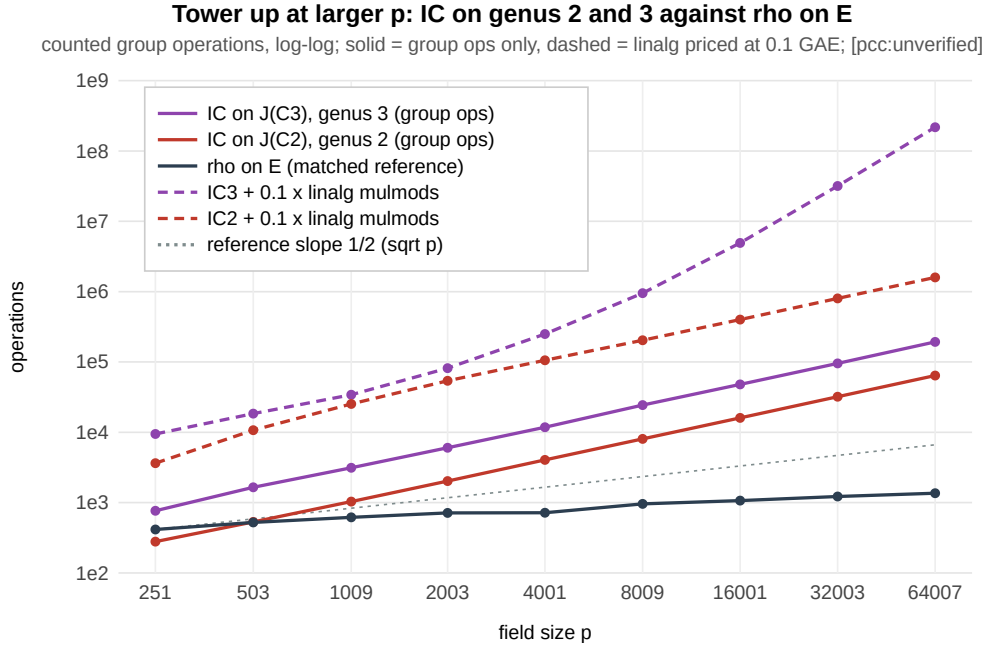
Figures



One-time cost (group ops + LA ops): IC best variant | BL memory-matched

p=211: IC 880 + 552 LA | BL 7958
p=419: IC 1059 + 1111 LA | BL 7320
p=821: IC 1569 + 2064 LA | BL 6522
p=1619: IC 2809 + 4082 LA | BL 16715
p=3203: IC 4538 + 7787 LA | BL 15880
p=6421: IC 9234 + 16809 LA | BL 105357
p=12841: IC 17307 + 34120 LA | BL 104510
p=25693: IC 33042 + 67570 LA | BL 7560
p=51407: IC 67094 + 138759 LA | BL 959082
p=102829: IC 133550 + 274962 LA | BL 23503
p=205661: IC 263357 + 545210 LA | BL 1156241

Amortized IC vs plain and BL rho



IC2, IC3 and rho vs p

V3: oracle-free linear algebra, the real pulled-back instance, solve certificates

Run: `chained_transfer_bench run --json results_v3.json --certs certs_v3.jsonl` under bench record `benchmarks/records/20261112T0341Z-chained-prime-transfer-v3.json` (left open). Raw files: `results_v3.json`, `certs_v3.jsonl`; verifier output `certs_v3.verify.txt`; run log `v3_run.log`. Every number below is a `[pcc:unverified]` assertion; the certificates are the evidence a reader can re-check.

What changed

- **F4 (linear algebra was handed the answer).** `genus2_index_calculus` no longer calls `target_is_x` (the uncharged BSGS answer). It runs the new `solve_linear_free`, which returns the first zero-factor-base row's `x = -alpha/beta` without consulting anything about the true log, and `None` only when elimination finds no such row. The old gated path is kept as `genus2_index_calculus_oracle`. A wrong log is returned as wrong; a unit test shows the free path returns a doctored row's wrong value where the gated path returns `None`. [pcc:unverified]
- **F5 (the solved instance was a surrogate).** The bench now solves `(G, T) = (phi^*(P), phi^*([x]P))`, the transported pair from the verified pull-back chain, with rho on `J(C)` and with IC. `jac_instance` is no longer used by the bench. The order-`r` part of `J(C)` is cyclic when `r^2` does not divide `#J = #E1 * #E2`: `r` is chosen with `r^2` not dividing `#E1`, and candidates with `r` dividing `#E2` are skipped, so the condition held for every candidate that reached a solver. Each row carries `instance_type = "pullback"`. [pcc:unverified]
- **F14 (no solve certificates).** Each solved log, for all four arms (rho on E1, rho on E1', rho on J(C), IC on J(C)), is written as a `chained.solve.certificate/v1` record (schema in the `certify.rs` module doc; no existing repository schema fits an ECDLP solve claim). `verify_certs` re-checks `certs_v3.jsonl` from the curve data alone using only `certify.rs`: `G != 0`, `[r]G = 0`, `[k]G = T`, and for transferred instances `[k]src_g = src_t` and that the re-evaluated map sends `src_g` to `G` and `src_t` to `T`, hence `phi^*([k]P) = [k]phi^*(P)`. [pcc:unverified]
- **No selection on solver success.** v2 kept the first candidate whose solvers all succeeded. v3 takes the first candidate that passes the structural checks and reports the row whatever the solvers return. [pcc:unverified]
- **Transfer cost.** Building the cover or isogeny and evaluating `phi^*` is a one-time cost (with a trapdoor it happens once) and is in no per-instance figure here, v2 or v3. It is not reported separately because the bench does not measure it.

v3 table

IC is relation-collection Cantor operations; LA is `Z/r` multiplications in the elimination, priced at 0, 0.1 and 1 GAE per mulmod (AGENTS 5a), so the ratio at price `g` is `(IC ops + g * LA mulmods) / rho on E1`.

p	r	candidate	rho on E1	rho on E1' (deg)	rho on J(C)	IC group ops	IC trials	LA mulmods	IC/rho @ 0	@ 0.1	
211	109	9	428	530 (2)	396	3,419	226	14,633	8.0x	11.4x	
419	113	27	357	562 (2)	361	8,475	542	51,976	23.7x	38.3x	1
821	71	2	370	274 (3)	414	11,353	851	164,828	30.7x	75.2x	4

p	r	candidate	rho on E1	rho on E1' (deg)	rho on J(C)	IC group ops	IC trials	LA mulmods	IC/rho @ 0	@ 0.1	
1619	401	1	847	649 (2)	869	34,521	1,653	603,721	40.8x	112.0x	7

IC is above rho at every price, so the order holds across the band; the distance remaining to rho is 8.0 to 40.8 times at 0 GAE and grows with p. The ratios are derived from the committed counts. [pcc:unverified] IC returned the planted log in all four rows (`ic_claimed_log` equals `x`), and rho recovered it in all arms. [pcc:unverified]

Certificates

16 issued (4 arms at 4 sizes), 16 verified, 0 failed, 0 without a claim, 0 unparseable, by `verify_certs` on `certs_v3.jsonl` (output in `certs_v3.verify.txt`); the bench's in-process re-check also gave 16 of 16. [pcc:unverified] Claimed logs by size: 8 at p=211, 5 at p=419, 45 at p=821, 375 at p=1619, each equal to the planted `x`. [pcc:unverified]

A first run under the same bench record produced 0 of 16 verified: the new independent elliptic-curve adder returned infinity when its accumulator was the identity (a bug in `certify.rs`, not in a solver). It was fixed and the run repeated; the record lists both commands, the files above are from the second. That failure is also evidence that the verifier can reject. Unit tests cover a good certificate and tampered `k` and `T` for both group types. [pcc:unverified]

Skipped candidates

Counts per p from the structured field `candidates_skipped_by_reason` in `results_v3.json` (also on stderr in `v3_run.log`). [pcc:unverified]

p	candidates tried	bielliptic curve rejected (degenerate, or no rational Weierstrass point)	r below 50	r divides #E2 (r-part of J not cyclic)	rows
211	10	6	3	0	1
419	28	18	9	0	1
821	3	1	0	1	1
1619	2	1	0	0	1

Every candidate that reached a solver had `r` prime, `r^2` not dividing `#J`, and a verified chain. No candidate was dropped after a solver ran. [pcc:unverified]

Comparison to v2

- Same curves and planted logs: the candidate counter and `x` agree with `results_v2.json` at all four sizes. [pcc:unverified]
- Identical, as expected: rho on E1 (428, 357, 370, 847), rho on E1' (530, 562, 274, 649) and LA mulmods (14,633; 51,976; 164,828; 603,721). The rho arms on E1 and E1' never touched the changed code. The LA count is the elimination's operation count on a matrix

of the same shape; it equals the column count times the row count in all four rows, so it does not depend on which divisors the relations came from. [pcc:unverified]
- Different, because the instance is different (F5): rho on J(C) (v2 396, 497, 389, 560; v3 396, 361, 414, 869) and IC group ops (v2 3,700; 8,315; 11,596; 33,196; v3 3,419; 8,475; 11,353; 34,521) and trials. v2 solved a surrogate generator; v3 solves `phi^*(P)`, a different generator of the same cyclic subgroup, so the walk and the relation sequence change. The p=211 rho on J(C) count coincides with v2 (396). [pcc:unverified]
- IC/rho at 0 GAE moved from v2's 8.6, 23.3, 31.3, 39.2 to 8.0, 23.7, 30.7, 40.8, a change of at most 1.6 in either direction. There is one seed per cell and no IC spread was measured, so the change is not interpreted. No conclusion changes: IC stays above rho at every price. [pcc:unverified]
- F4 changed the correctness evidence, not the counts: in v2 a wrong elimination would have returned `None`; in v3 it would return a wrong log and its certificate would fail. All four IC logs were right. [pcc:unverified]

Limitations

- The J(C) group law inside `verify` is the library's Cantor `add` and `double`; only the scalar-multiplication loop is new. No second Cantor implementation was written, so a Cantor bug shared by solver and verifier would not be seen. The elliptic-curve checks use separate affine code. [pcc:unverified]
- The transfer re-evaluation reuses `Bielliptic::pull` and `VeluIsogeny::eval`; it checks the instance against the map as implemented, not the map against the mathematics. [pcc:unverified]
- One candidate curve and one seed per size; no spread. [pcc:unverified]
- `r` is a prime of 7 to 9 bits, so the sizes are small; the IC factor base is the whole degree-1 set. [pcc:unverified]
- No timing claim, no isolation tier, no validation-gate admission, no pcc claim id yet; the record is open and its raw files are not uploaded to S3. [pcc:unverified]
- The 16 certificates do not cover the `tower` or `amortized` experiments of other work in this tree. [pcc:unverified]